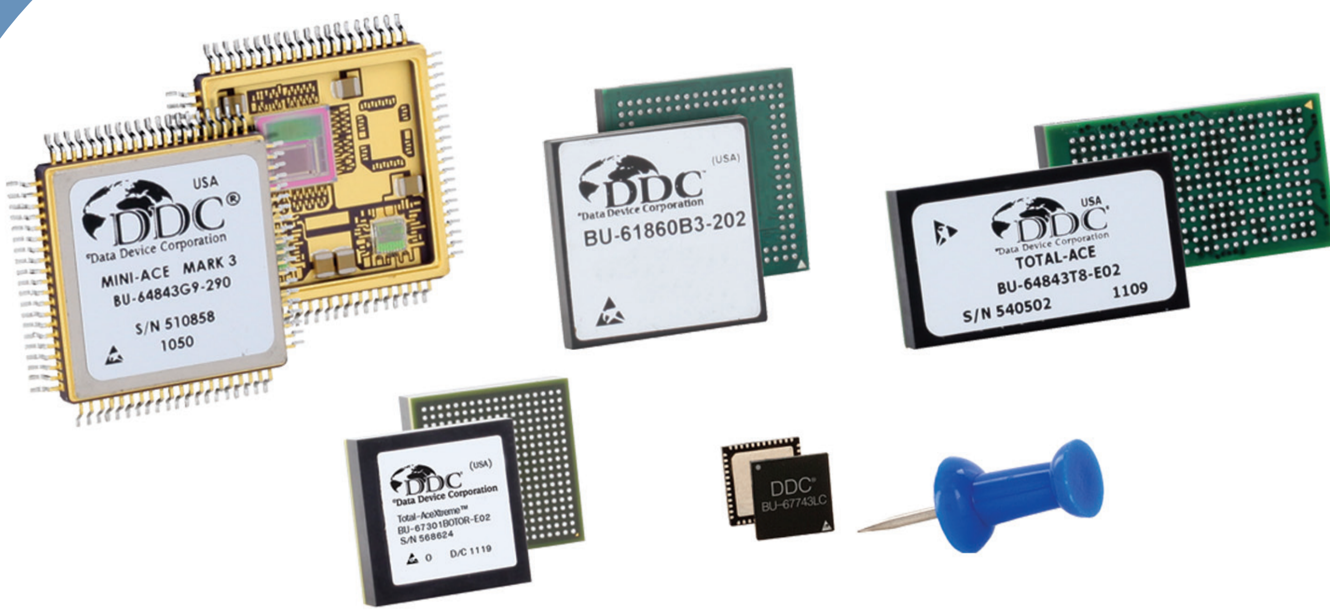
Host Interface Performance for MIL-STD-1553 Terminals

Application Note

AN/B-59



This document provides the information on MIL-STD-1553 terminals, the different host interfaces, and how the characteristics of the terminals vary between the different interfaces offered by DDC.

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1 INTRODUCTION

1.1 Purpose

MIL-STD-1553 terminals are available with several different host interfaces, from the classic local-bus variants to several different serial options to minimize signal count. Which interface is best for a particular application will be a trade-off between power, cost, layout complexity, and performance. This application note will take a look at several terminals that are offered by DDC and how some of these characteristics vary between the different interfaces that are offered.

1.2 Terminals

DDC offers a wide range of MIL-STD-1553 terminal components to suit various customer needs. These range from the fully integrated Total-Ace and Total-AceXtreme, which incorporate all necessary components including the isolation transformers, to the ceramic Mini-Ace Mark 3, to the 7mm x 7mm Nano-Ace.

Table 1. Component Options

Architecture	Terminal Name	Base Part Number(s)	Interface Options
AceXtreme	Total-AceXtreme	BU-67301B	Local-Bus or PCI
	PCIe-AceXtreme	BU-67302B	PCI-Express
Enhanced Mini-Ace (EMA)	Total-Ace	BU-64843T, BU-64863T, BU-64843U, BU-64863U, BU-64843H, BU-64863H, BU-64843I, BU-64863I	Local-Bus
	PCI Total-Ace	BU-65843T, BU-65863T, BU-65843U, BU-65863U, BU-65843H, BU-65863H, BU-65843I, BU-65863I	PCI
	Mini-Ace Mark 3	BU-64745F, BU-64845F, BU-64743F, BU-64843F, BU-64863F, BU-64745G, BU-64845G, BU-64743G, BU-64843G, BU-64863G	Local-Bus
	PCI Mini-Ace Mark 3	BU-65743F, BU-65843F, BU-65863F, BU-65743G, BU-65843G, BU-65863G	PCI
	Micro-Ace TE	BU-64840B, BU-64860B, BU-64843B, BU-64863B, BU-64840R, BU-64860R, BU-64843R, BU-64863R	Local-Bus
	PCI Micro-Ace TE	BU-65843B, BU-65863B, BU-65864B BU-65843R, BU-65863R, BU-65864R	PCI
	Nano-Ace	BU-67743L	SPI
EMA or Flex-Core	IP	BU-692XX	Transceiver

2 TRADE-OFFS

As integration levels steadily increase with more functions being packed into each box, board, FPGA or Controller, space is an increasingly valuable commodity. This affects terminal choice in two ways. First, there is the board area (and in some cases, volume) occupied by the 1553 components. Second, there is the I/O interface with the host, and the tradeoff between pin count, complexity, and performance.

Two other aspects that may be either critical or meaningless depending on the design are software compatibility and certifiability. DDC's EMA devices are all compatible with software created for other EMA devices, as well as DDC's older Ace and Mini-Ace devices, and with minimal changes to the AIM series devices as well. Because of the number of existing applications and extensive service history, DDC is also able to provide the documents and artifacts required to certify applications using local-bus EMA components to any level of DO-254.

AceXtreme devices support a new software interface that allows much greater performance, and which is architecturally similar to Enhanced Mini-Ace devices, but not register compatible. However, DDC also provides the BU-69092 AceXtreme C Software Development Kit (SDK), which will natively support PCI and PCI-Express devices using both architectures (as well as standard PCI, PCI-Express, and USB boards and modules available from DDC) without any application code changes. Supported operating systems include Windows, Linux, VxWorks, and Integrity. A Driver Development Kit (DDK) with full driver source code is also available to allow support for other devices.

2.1 Board Area

There are several options to decrease board area requirements. The obvious first choice is to integrate everything into as few, small packages as possible. Where once a whole computer board might contain a single terminal interface made up of many simple analog and digital components, integrated transceiver and protocol chips have been in use for some time now. More recently, DDC has released the Total-Ace and Total-AceXtreme which take the next logical step by integrating the entire terminal (protocol logic, memory, transceiver, and isolation transformer) into a single BGA package.

Some systems take a slightly different route and integrate the memory and protocol logic into the system FPGA, or even an SoC ASIC, leaving the analog transceiver and transformer to be placed separately on the PCB. This technique is generally less advantageous than it may appear at first glance, as it increases the size of the required FPGA, increasing the cost and frequently the board area as well. In addition, integrating part of the 1553 includes the extra task of passing the full RT or BC Validation testing, as well as system safety certifications. These can add significantly to the development cost of a program, both in money and time to completion.

In the table below, those terminal solutions that do not integrate the required isolation transformer are paired with a TSM-2230 from Beta Transformer Technology Corporation (BTTC). This is one of the smallest dual transformer packages on the market.

Table 2. Estimated Board Area Used

Terminal	Board Area	Notes
Total-AceXtreme	~280mm^2	
PCIe-AceXtreme	~485mm^2	w/TSM-2230
Total-Ace	~530mm^2	
PCI Total-Ace	~530mm^2	
Mini-Ace Mark 3	~1000mm^2	w/TSM-2230
PCI Mini-Ace Mark 3	~1000mm^2	w/TSM-2230
Micro-Ace TE	~625mm^2	w/TSM-2230
PCI Micro-Ace TE	~625mm^2	w/TSM-2230
Nano-Ace	~220mm^2	w/TSM-2230
IP + Transceiver	~220mm^2 + FPGA	w/TSM-2230, extra FPGA space not included

The Nano-Ace and the BU-67401 transceiver can also be placed on the bottom of the PCB under the transformer, reducing the total area used (both sides of PCB) to 160mm^2.

2.2 Signal Count

One side effect of the trend towards greater integration has been to increase the pressure on the IO interface pins. As more functions are packed into a smaller package, the required interface bandwidth increases, just as the availability of I/O pins becomes constrained. This has lead to the development and increasing popularity of serial interfaces, with options ranging in performance and complexity from PCI-Express, which will provide equal or greater performance compared to standard PCI, to SPI, which offers simple design and minimal hardware requirements. At the same time, there remain many applications where the ease of implementation and minimal latency of the classic local-bus are worth the cost of the extra pins; while IP provides the options of a fully custom interface, the moderate number of required I/O pins to connect to the external transceiver is more than for an SPI or PCI-Express interface.

Table 3. Host Interface Signal Count	
Terminal	**Interface Signal Count**
Total-AceXtreme (Local-Bus)	21-56
Total-AceXtreme (PCI)	48
PCIe-AceXtreme	6
Total-Ace	20-36
PCI Total-Ace	46
Mini-Ace Mark 3	20-36
PCI Mini-Ace Mark 3	46
Micro-Ace TE	20-36
PCI Micro-Ace TE	46
Nano-Ace	4
IP + Transceiver	10

2.3 Performance

While the maximum bandwidth of the Mil-Std-1553 interface is fixed by the 1 MHz operating frequency at ~970 Kb/s and terminals typically operate well below that, latency requirements and host processor overhead will be affected by the performance of the interface to the terminal. IP can support multiple interface options, including internal dual-port ram of arbitrary width within the FPGA, so there are no inherent performance limitations.

Please note that the numbers in the tables below assume a 20MHz operating clock for the Total-Ace and uncontended accesses. Access latencies for the PCI and Local-Bus versions of the Total-Ace, Mini-Ace, and Micro-Ace may be increased due to bus contention with the protocol logic. The AceXtreme and Nano-Ace components use a pseudo dual-port architecture to eliminate bus contention. Local-bus versions are also buffered and not multiplexed and operated at their maximum width (16-bit for Total-Ace etc. and 32-bit for Total-AceXtreme). Total-AceXtreme is operated in synchronous mode with an 80 MHz clock. The SPI port on Nano-Ace is operated at 50 MHz.

Table 4. Host Bus Performance		
Terminal	**Bandwidth**	**Read Latency (First Word)**
Total-AceXtreme (Local-Bus)	320 MB/s	62.5ns
Total-AceXtreme (PCI)	266 MB/s	165ns
PCIe-AceXtreme	250 MB/s	165ns + latency of internal PEX-8112 PCIe to PCI bridge
Total-Ace	6 MB/s	270ns

Table 4. Host Bus Performance		
Terminal	**Bandwidth**	**Read Latency (First Word)**
PCI Total-Ace	33 MB/s write, 3.4 MB/s read	600ns
Mini-Ace Mark 3	6 MB/s	270ns
PCI Mini-Ace Mark 3	33 MB/s write, 3.4 MB/s read	600ns
Micro-Ace TE	6 MB/s	270ns
PCI Micro-Ace TE	33 MB/s write, 3.4 MB/s read	600ns
Nano-Ace	6.25 MB/s (50 Mb/s)	960ns
IP	N/A	N/A

We can get a more realistic measurement of performance by calculating the amount of time that will be required to handle a message in BC or RT mode. Table 5 below shows the time used to support a single message transfer across the interface between the terminal component and the host for polled operation (to remove the system and OS dependent aspects of interrupt latency) and leaves out the time required for host processor operations and host memory accesses since those can vary significantly between systems.

To handle a BC message, it is assumed that the host polls the BC Block Status word for the next message to be processed, finds that it is not complete, polls it again, sees that it is complete, then clears the status and either reads (TX command) or writes the corresponding data (RX command).

To handle an RT message it is assumed that the host polls the Interrupt Status register, gets nothing, polls again and gets an EOM, then reads the next Message Block from the Command Stack, and follows the Data Pointer to either write (TX command) or read the corresponding data data (RX command).

Table 5 below shows cases for BC, RT, TX, RX and 1 or 32 words of data. Once again, IP provides too many options for customization to allow a meaningful number to be compared here.

Table 5. Host Bus Time per Message (us)								
Terminal	**BC**				**RT**			
	TX		**RX**		**TX**		**RX**	
	1	**32**	**1**	**32**	**1**	**32**	**1**	**32**
Total-AceXtreme (Local-Bus)	0.6875	0.875	0.575	0.7625	0.725	0.9125	0.8375	1.025
Total-AceXtreme (PCI)	0.780	1.005	0.780	0.9	0.825	0.945	0.825	1.05

Table 5. Host Bus Time per Message (us)

Terminal	BC				RT			
	TX		RX		TX		RX	
	1	32	1	32	1	32	1	32
Total-Ace	1.190	10.8	1.140	9.2	2.12	10.18	2.17	11.78
PCI Total-Ace	1.920	20.2	1.440	2.79	3.68	5.03	4.16	22.4
Mini-Ace Mark 3	1.190	10.8	1.140	9.2	2.12	10.18	2.17	11.78
PCI Mini-Ace Mark 3	1.920	20.2	1.440	2.79	3.68	5.03	4.16	22.4
Micro-Ace TE	1.190	10.8	1.140	9.2	2.12	10.18	2.17	11.78
PCI Micro-Ace TE	1.920	20.2	1.440	2.79	3.68	5.03	4.16	22.4
Nano-Ace	N/A	N/A	N/A	N/A	4.48	14.56	4.8	14.72
IP	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

For components that support burst access (Total-AceXtreme, PCIe AceXtreme, Nano-Ace, and the other PCI components for writes), performance can be improved even further if the system configuration allows the data from multiple messages to be accessed with a single transfer. To this end, the two AceXtreme components also support DMA.

A 1553 message consists of a command word from the BC, a status word from the RT, and up to 32 data words. Assuming a RT response latency of 10us and a 10us gap time, that means that a 1DW message will take 80us to transmit on the 1553 bus, while a 32 DW message will require 700us. Recalculating the numbers in Table 5 to show how much of the available bus time is used to service the 1553 terminal yields Table 6 below.

Table 6. Host Bus Utilization (%)

Terminal	BC				RT			
	TX		RX		TX		RX	
	1	32	1	32	1	32	1	32
Total-AceXtreme (Local-Bus)	0.86	0.13	0.72	0.11	0.91	0.13	1.0	0.15
Total-AceXtreme (PCI)	0.98	0.14	0.98	0.13	1.0	0.14	1.0	0.15
Total-Ace	1.5	1.5	1.4	1.3	2.7	1.5	2.7	1.7
PCI Total-Ace	2.4	2.9	1.8	0.40	4.6	0.72	5.2	3.2
Mini-Ace Mark 3	1.5	1.5	1.4	1.3	2.7	1.5	2.7	1.7
PCI Mini-Ace Mark 3	2.4	2.9	1.8	0.40	4.6	0.72	5.2	3.2
Micro-Ace TE	1.5	1.5	1.4	1.3	2.7	1.5	2.7	1.7
PCI Micro-Ace TE	2.4	2.9	1.8	0.40	4.6	0.72	5.2	3.2

Table 6. Host Bus Utilization (%)								
Terminal	BC				RT			
	TX		RX		TX		RX	
	1	32	1	32	1	32	1	32
Nano-Ace	N/A	N/A	N/A	N/A	5.6	2.1	6.0	2.1
IP	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

3 APPLICATIONS

Choosing the appropriate terminal component is a matter of weighing the requirements of the application against the various features of the devices and choosing the one that comes closest to matching the most critical aspects. Some systems will need the reliability to survive in harsh environments and the resilience to shock, vibration, and temperature cycling that is provided by a hermetically-sealed, ceramic case and gull-wing leads, for which the Mini-Ace Mark 3 and PCI Mini-Ace Mark 3 become the obvious candidates. In most cases the smaller size, reduced cost, and potential for higher performance or reduced signal count will push the choice to one of the other options.

Here are some examples that show some of the range of applications for 1553.

3.1 Mission Computer

This application serves as the controller of a 1553 bus, and thus must manage the scheduling and data for every message is transmitted across the bus. Typically a relatively high performance processor is present, but it's also important not to tie it up dealing with the I/O interface as the performance is usually required for other, compute-intensive tasks. Size and weight are also important. In many cases, there will be multiple 1553 interfaces.

The CPU and/or chipset in this system will probably have a few PCI or PCI-Express ports built in, so the Total-AceXtreme and PCIe AceXtreme are attractive options. However, because PCIe buses are point-to point, if multiple 1553 interfaces are required, then a PCIe switch would be needed to connect multiple PCIe AceXtreme components. Given that a bridge will be needed anyway, a PCI-Express to PCI bridge will allow the connection of up to eight Total-AceXtreme components. If only a single 1553 channel is needed, the PCIe-AceXtreme may be chosen to save the bridge, but it will require the addition of the 1553 isolation transformer.

The fast host bus interface, built-in DMA controller, and powerful BC engine mean that even with multiple channels in operation, the load on the host CPU will minimal, allowing it to concentrate on processing data rather than merely transferring it.

3.2 Simple RT

This is a very simple embedded computer that has a limited number of functions: control simple actuators, and/or report sensor data over the 1553 bus. It does not require high performance, so it uses a very simple microcontroller that does not include even an external memory bus, let alone advanced buses like PCI and PCI-Express. Many of these controllers, however, do include built-in SPI ports.

Even if hardware SPI is not available, it can be implemented in software very easily using four general purpose I/O signals, and while there is a performance penalty for doing so, the BC only interrogates this RT once per frame, so there is plenty of time to process each request and prepare for the next one.

This is an application where cost, space, pin-count, and ease of interface all point to Nano-Ace as the solution.

3.3 Civilian Flight Control System

A customer was looking to consolidate flight control wiring into a bus and settled on MIL-STD-1553 for its proven record of reliability and noise immunity. Because of the safety-critical nature of this application, the System Safety Assessment designated the subsystem as Level A under both DO-178 for the software and DO-254 for the hardware. DDC is able to supply the necessary documentation to support DO-254 certification up to Level A for the Mini-Ace Mark 3, Micro-Ace TE and Total-Ace.

The Flight Control System is actually made up of multiple LRUs, placed in different locations in the aircraft. As a result, the environmental requirements vary, so the choice between those components was made differently for individual LRUs. Cockpit controls can save space and/or cost by going with the Total-Ace or Micro-Ace TE, while sensors and actuators that are subject to extreme temperature swings or higher shock may require the Mini-Ace Mark 3.

Table 7. Applications

Application	Best Fit	Reasons
Mission Computer	Total-AceXtreme PCIe-AceXtreme	Compute Intensive Minimizes CPU load Minimizes size/weight requirement PCI/PCIe interface easily integrates into system

Table 7. Applications

Application	Best Fit	Reasons
Tech Refresh	Total-Ace	Fully integrated terminal • Occupies much less space • Simplifies circuit design Fully compatible with legacy software
Remote Sensor	Nano-Ace	Size/weight at a premium Low duty cycle on 1553 bus leads to low bandwidth/processing requirement at host interface Simple microcontroller used to collect and process sensor data has an integrated SPI port, but no external memory interface or high-performance peripheral ports.
Flight Control System	Total-Ace	DO-254 Certification Support Package Fully integrated terminal • Occupies much less space • Simplifies circuit design
Engine Control System	Mini-Ace Mark 3	DO-254 Certification Support Package Extreme environmental requirements
Missile	Mini-Ace Mark 3	Extreme environmental requirements

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